# Econometria

## Esercizi originali con soluzioni

Cinque problem set originali, ciascuno seguito dal relativo fascicolo di soluzioni.

Non sono stati trovati appelli o prove generate per questo blocco di econometria: il volume e' denominato esercizi per distinguerlo dagli esami.

**PROBLEM SET 1**

1) Consider the linear regression model: $y = X\beta + u$. Suppose that all the assumptions made in class hold, in particular $u \sim N(0, \sigma^2 I)$ and $E(u|X)=0$.

(a) Show that the OLS estimator b is unbiased: $E(b) = \beta$.

(b) Derive the conditional variance of b.

(c) How can we estimate the variance of the OLS estimator derived in (b)?

2) Consider the classical linear regression model: $y = X\beta + u$.
   a) Derive the coefficient of determination, assuming the regression includes a constant, and discuss why this hypothesis is important
   b) Discuss potential problems with the coefficient of determination and how they can be addressed

3) Consider the following linear regression model: $y_i = \beta_0$. Derive the least squares estimator of the parameter in the model and discuss the result.

4) Consider the linear regression model: $y = X\beta + u$. Suppose that all the assumptions made in class hold, in particular $u \sim N(0, \sigma^2 I)$ and $E(u|X)=0$.

   (a) Write down a test for the significance of a subset J of regressors using the residual sum of squares of the general and of the restricted model. How is the test statistic distributed?

   (b) Show how the test in (a) can be written in terms of the $R^2$ from the general and the restricted regressions.

5) Consider the linear regression model: $y = X\beta + u$. Suppose that all the assumptions made in class hold, in particular $u \sim N(0, \sigma^2 I)$ and $E(u|X)=0$. We are interested in testing a set of J linear restrictions on the parameter vector:

$$H_0: \boldsymbol{R}\beta = r$$

where **R** is a J×K matrix and r is a J×1 vector.

   a) Show how we can write an F statistic to test the above set of hypothesis

   b) Assume that k=3 and that you want to test jointly whether $\beta_2 = 3$ and $\beta_3 = 1 - \beta_1$. Show how to write the elements of the matrix **R** and of the vector r in this case.

**PROBLEM SET 1**

1) Consider the linear regression model: $y = X\beta + u$. Suppose that all the assumptions made in class hold, in particular $u \sim N(0, \sigma^2 I)$ and $E(u|X)=0$.

(a) Show that the OLS estimator b is unbiased: $E(b) = \beta$.

(b) Derive the conditional variance of b.

(c) How can we estimate the variance of the OLS estimator derived in (b)?

*Solution*: *see slides set 1 part 2, pp. 1-3 and 6-7*

2) Consider the classical linear regression model: $y = X\beta + u$.
   a) Derive the coefficient of determination, assuming the regression includes a constant, and discuss why this hypothesis is important
   b) Discuss potential problems with the coefficient of determination and how they can be addressed

*Solution*: *see slides set 1 part 1, pp. 36-40*

3) Consider the following linear regression model: $y_i = \beta_0$. Derive the least squares estimator of the parameter in the model and discuss the result.

*Solution: solve* $min \sum_i (y_i - b_0)^2$ *by taking the first derivative w.r.t.* $b_0$ *and get* $\sum_i y_i - nb_0 = 0$*, hence the least squares estimator is the sample mean.*

4) Consider the linear regression model: $y = X\beta + u$. Suppose that all the assumptions made in class hold, in particular $u \sim N(0, \sigma^2 I)$ and $E(u|X)=0$.

   (a) Write down a test for the significance of a subset J of regressors using the residual sum of squares of the general and of the restricted model. How is the test statistic distributed?

   (b) Show how the test in (a) can be written in terms of the $R^2$ from the general and the restricted regressions.

*Solution*: *see slides set 1 part 2, pp. 14-15*

5) Consider the linear regression model: y = Xβ +u. Suppose that all the assumptions made in class hold, in particular u ~ N(0, $\sigma^2$I) and E(u|X)=0. We are interested in testing a set of J linear restrictions on the parameter vector:

$$H_0: \boldsymbol{R}\beta = r$$

where **R** is a J×K matrix and r is a J×1 vector.

*a)* Show how we can write an F statistic to test the above set of hypothesis

*Solution: slides part 1 set 2, pp. 20-22*

b) Assume that k=3 and that you want to test jointly whether $\beta_2 = 3$ and $\beta_3 = 1 - \beta_1$. Show how to write the elements of the matrix **R** and of the vector r in this case.

*Solution:* $R = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad r = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

**PROBLEM SET 2**

1) Consider the regression output:

| Source | SS | df | MS | Number of obs | = | 6,070 |
|---|---|---|---|---|---|---|
| | | | | F(3, 6066) | = | 95.66 |
| Model | 327210.935 | 3 | 109070.312 | Prob > F | = | 0.0000 |
| Residual | 6916280.95 | 6,066 | 1140.1716 | R-squared | = | 0.0452 |
| | | | | Adj R-squared | = | 0.0447 |
| Total | 7243491.89 | 6,069 | 1193.52313 | Root MSE | = | 33.766 |

| rate | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| age | 2.209225 | .2049644 | 10.78 | 0.000 | 1.807422 | 2.611028 |
| age2 | -.0213555 | .002253 | -9.48 | 0.000 | -.0257722 | -.0169389 |
| female | -9.125352 | .8675678 | -10.52 | 0.000 | -10.82609 | -7.424611 |
| _cons | -16.58503 | 4.384512 | -3.78 | 0.000 | -25.18023 | -7.989829 |

where rate is the hourly wage rate in US dollars in the US in the year 2019, age is the age of the individual, age2 is the age squared, and female is a dummy variable equal to 1 if the individual is a woman and zero otherwise.

a) What is the effect on the wage rate of an increase in age of 1 year? Comment briefly.

b) What is the predicted wage rate for a 40 years old men?

c) What can we say about the difference in the wage rate between men and women in the sample?

d) Compute the statistic to test that all the variables in the regressions have coefficients equal to zero: write both the formula and the result.

2) Consider the following regression output, where *lrate* is the natural logarithm of the hourly wage rate, *age* is age of the individual, *nchild* is the number of children in the household, *female* is a dummy equal to 1 if the individual is a woman and zero otherwise, and *nchild_female* is the product of the number of children and the dummy *female*.

| Source | SS | df | MS | Number of obs | = | 6,047 |
|---|---|---|---|---|---|---|
| | | | | F(4, 6042) | = | 71.72 |
| Model | 166.618408 | 4 | 41.654602 | Prob > F | = | 0.0000 |
| Residual | 3509.23844 | 6,042 | .580807421 | R-squared | = | 0.0453 |
| | | | | Adj R-squared | = | 0.0447 |
| Total | 3675.85685 | 6,046 | .607981615 | Root MSE | = | .76211 |

| lrate | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| age | .0072859 | .0007604 | 9.58 | 0.000 | .0057952 | .0087766 |
| nchild | .0747049 | .0114552 | 6.52 | 0.000 | .0522486 | .0971611 |
| female | -.1686385 | .0245204 | -6.88 | 0.000 | -.2167073 | -.1205697 |
| nchild_female | -.0820583 | .016163 | -5.08 | 0.000 | -.1137435 | -.0503732 |
| _cons | 2.856763 | .0379752 | 75.23 | 0.000 | 2.782318 | 2.931208 |

1) What is the estimated effect of age on the wage rate?

2) What is the estimated effect of having one child on the wage? Is this effect the same for men and women?

3) Compute the difference in log rate (*lrate*) between a man and a woman both aged 40 with one child is (in absolute value):

4) Test the significance of the coefficient on the interaction between *female* and *nchild* at $\alpha = 0.05$ (critical value=1.96). The answer must include (i) the null hypothesis, (ii) the alternative hypothesis, (iii) the formula of the test statistic and its distribution, (iv) the rejection rule and (v) the conclusion of the test (vi) the interpretation of the result

**Question 3**
Assume the true model is y = Xβ+Zγ +u , but instead you estimate the restricted model y = Xβ +u. Is the estimated OLS parameter vector an unbiased estimate of the true β? Explain, assuming u ~ $N(0, \sigma^2 I)$ and that all explanatory variables are exogenous.

**Question 4** The partitioned regression

Suppose that a regression involves two sets of variables, X1 and X2:

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon} = \boldsymbol{X_1\beta_1} + \boldsymbol{X_2\beta_2} + \boldsymbol{\varepsilon}$$

What is the algebraic solution for $\mathbf{b_2}$? How can we interpret it?

**Question 5** The partitioned regression

Suppose that a regression involves only one variable, x, in addition to a constant:

$$y = \beta_1 + x\beta_2 + \varepsilon$$

Use the partitioned regression results to show what is the algebraic solution for $\mathbf{b_2}$. How can we interpret it?

**PROBLEM SET 2**

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where rate is the hourly wage rate in US dollars in the US in the year 2019, age is the age of the individual, age2 is the age squared, and female is a dummy variable equal to 1 if the individual is a woman and zero otherwise.

a) What is the effect on the wage rate of an increase in age of 1 year? Is it the same at age 20 or at age 40? Comment briefly.

*Because age enters with a quadratic term, the effect of an increase in age depends on age as it is: b_age+2*b_age2*age=2.21-2*0.021*age*
*Hence, for example, at age 20 it is = 1.36 , while at age 40 it is smaller and = 0.506*

b) What is the predicted wage rate for a 40 years old men?

*E(rate|female=0, age=40)=-16.585+2.21*age-0.021*age*age = 38.215*

c) What can we say about the difference in the wage rate between men and women in the sample?

*Conditional on age (i.e. keeping age constant, i.e. comparing a men and a woman of the same age), women earn a wage rate 9.13 dollars lower than man.*

d) Compute the statistic to test that all the variables in the regressions have coefficients equal to zero: write both the formula and the result.

*The simplest formula for this F test is* $F = \frac{R^2/(k-1)}{(1-R^2)/(n-k)}$ *where the R^2 refers to our estimated equation (so it is 0.0452). This version of the test uses the fact that the R^2 for the restricted model is zero.*

2) Consider the following regression output, where *lrate* is the natural logarithm of the hourly wage rate, *age* is age of the individual, *nchild* is the number of children in the household, *female* is a dummy equal to 1 if the individual is a woman and zero otherwise, and *nchild_female* is the product of the number of children and the dummy *female*.

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| lrate | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
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| female | -.1686385 | .0245204 | -6.88 | 0.000 | -.2167073 | -.1205697 |
| nchild_female | -.0820583 | .016163 | -5.08 | 0.000 | -.1137435 | -.0503732 |
| _cons | 2.856763 | .0379752 | 75.23 | 0.000 | 2.782318 | 2.931208 |

1) What is the estimated effect of age on the wage rate?

*Solution: The wage rate is in logarithms, age is in levels. Hence, the estimated increase in the wage rate following a 1 unit increase in age is about 0.7%*

2) What is the estimated effect of having one child on the wage? Is this effect the same for men and women?

*The effect is precisely estimated and indicates that for men the effect of every child is positive (+7.5% in wage rate approximately). For women, the effect is reversed (about 7.4%-8.2%=-0.8% for each child)*

3) Compute the difference in log rate (*lrate*) between a man and a woman both aged 40 with one child is (in absolute value):

*Solution: 0.25*

4) Test the significance of the coefficient on the interaction between *female* and *nchild* at $\alpha = 0.05$ (critical value=1.96). The answer must include (i) the null hypothesis, (ii) the alternative hypothesis, (iii) the formula of the test statistic and its distribution, (iv) the rejection rule and (v) the conclusion of the test (vi) the interpretation of the result

*Answer: H0: beta=0, H1: beta!=0; test statistic is (beta_hat-0)/SE(beta_hat), distributed as t or normal; reject if t in absolute value>1.96; rejected. Intepretation: difference in effect of having*

*children on wage rate between men and women.*
*The coefficient in the interaction tells us that the effect of children on the wage rate is lower for women than for men.*

**Question 1**
Assume the true model is y = Xβ+Zγ +u , but instead you estimate the restricted model y = Xβ +u. Is the estimated OLS parameter vector an unbiased estimate of the true β? Explain, assuming u ~ $N(0, \sigma^2 I)$ and that all explanatory variables are exogenous.

*Answer: Slides part1 set 3, slides 21-23*

**Question 2**
The partitioned regression

Suppose that a regression involves two sets of variables, X1 and X2:

$$\boldsymbol{y = X\beta + \varepsilon = X_1\beta_1 + X_2\beta_2 + \varepsilon}$$

What is the algebraic solution for $\mathbf{b_2}$? How can we interpret it?

*Answer: Slides part 1 set 3, slides 17-19*

**Question 3** The partitioned regression

Suppose that a regression involves only one variable, x, in addition to a constant:

$$y = \beta_1 + x\beta_2 + \varepsilon$$

Use the partitioned regression results to show what is the algebraic solution for $\mathbf{b_2}$. How can we interpret it?

*Answer: Slides part 1 set 3, slides 17-19. In this simple case, this is just the ols estimate of the regression run on demeaned y and demeaned X*

**Problem set 3**

**Question 1 GLS**

Consider the linear regression model: y = Xβ +u. Suppose that E(u|X)=0, but the disturbances are non-spherical, with $V(\varepsilon|X) = \sigma^2\Omega$, where Ω is a positive definite matrix.

a) Show whether the OLS estimator is unbiased
b) What is the variance/covariance matrix of the OLS estimator? How does this compare to the variance of the OLS estimator under the assumption of homoskedasticity?
c) If the error terms are heteroskedastic but not autocorrelated, what form does the Ω matrix have? In this case, can you suggest an estimator for the var/cov matrix in point b)?
d) Assume Ω is known, can you propose an efficient estimator? Derive this efficient estimator, and derive its mean and its variance.

**Question 2**

Given the linear model with K variables:

$$y_i = \boldsymbol{x}_{1i}'\boldsymbol{\beta_1} + x_{2i}\beta_2 + \varepsilon_i$$

where $\boldsymbol{x}_{1i}'$ is a vector of (K-1) exogenous variables, and $x_{2i}$ is one endogenous variable. You also observe an instrumental variable $z_{2i}$.

(a) write the moment conditions for the K-1 exogenous variables in $\boldsymbol{x}_{1i}'$ and for the endogenous variable $x_{2i}$, using the instrumental variable available to you
(b) derive the IV estimator for β, defining the (n x K) matrix Z as the matrix containing all the (K-1) exogenous variables + the instrument z.
(c) write the conditions under which the variable $z_{2i}$ is a valid instrument and discuss
(d) state whether the IV estimator is consistent and write its asymptotic distribution
(e) It is said that the standard errors of the IV estimator are typically bigger than the corresponding OLS standard errors. Can you give the intuition about why this is the case?

**Question 3**

Given the linear model with $R \geq K$ moment conditions:

$$E(\boldsymbol{z_i}\varepsilon_i) = \mathbf{0}$$
$$\varepsilon_i = y_i - x_i'\beta$$

(a) write the relevant quadratic form to be minimized in the GMM framework;
(b) derive the corresponding GMM estimator for β
(note that $\frac{1}{n}\sum_i[\boldsymbol{z}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})]$ can be also written as $\frac{1}{n}\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X\beta})$) ;
(c) write the expression for the *optimal* GMM estimator for β;
(d) show that the GMM estimator derived in (b) does not depend on the choice of the weighting matrix when R = K
(e) state under which conditions the GMM estimator derived in (b) is the OLS estimator.

## Question 4

The following table is the first step in a 2SLS estimation of the effect of education on earnings (from Card 1995).

```
Number of obs =  3,010
F(6, 3003)    = 608.02
Prob > F      = 0.0000
R-squared     = 0.4745
Adj R-squared = 0.4734
Root MSE      = 1.9425
```

| ed76 | Coefficient | Robust std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| exp | -.4100081 | .0319791 | -12.82 | 0.000 | -.4727113 | -.3473049 |
| exp2 | .0732287 | .1702908 | 0.43 | 0.667 | -.2606697 | .4071272 |
| black | -1.006138 | .0877849 | -11.46 | 0.000 | -1.178263 | -.8340136 |
| reg76r | -.291464 | .0784454 | -3.72 | 0.000 | -.4452761 | -.1376519 |
| smsa76r | .4038769 | .0849467 | 4.75 | 0.000 | .2373173 | .5704364 |
| nearc4 | .3373208 | .0806044 | 4.18 | 0.000 | .1792753 | .4953663 |
| _cons | 16.65917 | .1467408 | 113.53 | 0.000 | 16.37145 | 16.9469 |

Where the variable “ed76” is education, exp and exp2 are experience and experience squared (over 100), black is a dummy variable equal to one if the person is black, reg76r is a dummy for living in the south and smsa76r is a dummy for urban area. Finally, nearc4 is the instrument and it is a dummy = 1 if the person grew up near 4-yr college.

a - what is considered the endogenous variable in this setting? What is the exogenous instrument? What are the other variables included in the first stage regression?

b – How can we test if the instrument is weak? What are the results in the example?

c – how can we test is the instrument is really exogenous, i.e. if it does not enter the main equation explaining earnings as a function of education and other covariates?

## Question 5

After running a 2SLS estimation of a model of earnings on education and other variables (Card 1995), you run the following test:

```
. estat overid

  Test of overidentifying restrictions:

  Score chi2(1)          =  .869261  (p = 0.3512)
```

In your model, you considered education as the only endogenous variable.

Explain what is a test of overidentifying restrictions in this context and how you would interpret it. How many instruments did you use in the estimation of your model?

**Problem set 3**

**Question 1 GLS**

Consider the linear regression model: y = Xβ +u. Suppose that E(u|X)=0, but the disturbances are non-spherical, with $V(\varepsilon|X) = \sigma^2\Omega$, where Ω is a positive definite matrix.

a) Show whether the OLS estimator is unbiased
b) What is the variance/covariance matrix of the OLS estimator? How does this compare to the variance of the OLS estimator under the assumption of homoskedasticity?
c) If the error terms are heteroskedastic but not autocorrelated, what form does the Ω matrix have? In this case, can you suggest an estimator for the var/cov matrix in point b)?
d) Assume Ω is known, can you propose an efficient estimator? Derive this efficient estimator, and derive its mean and its variance.

*Answer:*

*a) Slides part1_set4 p. 3*
*b) Slides part1_set4 p. 4*
*c) It is a diagonal matrix. Can be estimated with White estimator (p. 19-20)*
*d) Slides p. 7-10 (formula 16 for the GLS estimator)+p.11 for unbiasedeness+p.12 for the variance*

**Question 2**

Given the linear model with K variables:

$$y_i = \boldsymbol{x}'_{1i}\boldsymbol{\beta_1} + x_{2i}\beta_2 + \varepsilon_i$$

where $\boldsymbol{x}'_{1i}$ is a vector of (K-1) exogenous variables, and $x_{2i}$ is one endogenous variable. You also observe an instrumental variable $z_{2i}$.

(a) write the moment conditions for the K-1 exogenous variables in $\boldsymbol{x}'_{1i}$ and for the endogenous variable $x_{2i}$, using the instrumental variable available to you
(b) derive the IV estimator for β, defining the (n x K) matrix Z as the matrix containing all the (K-1) exogenous variables + the instrument z.
(c) write the conditions under which the variable $z_{2i}$ is a valid instrument and discuss
(d) state whether the IV estimator is consistent and write its asymptotic distribution
(e) It is said that the standard errors of the IV estimator are typically bigger than the corresponding OLS standard errors. Can you give the intuition about why this is the case?

*Answers:*

*(a) Slides part2 set2 p. 10-11; in matrix form $Z'(y - X\widehat{\beta_{IV}})$=0, where X contains all X variables (endogenous and exogenous). Notice that here x_1 is a vector of K-1 variables instead of a single variable as in the slides.*
*(b) Slides part2_set2 p. 12*

(c) *Slides part2 set2 p.13 eq. 10: the instruments are orthogonal to the error terms and p. 14: the first condition is "relevance", i.e. the instruments are correlated with the endogenous variables*
(d) *Slides part2_set2 p. 16*
(e) *If the correlation between $x_2$ and $z_2$ is small (captured by matrix $\Sigma_{XZ}$) then the variance of the estimator is big, because its inverse enters the variance/covariance matrix*

**Question 3**

Given the linear model with $R \geq K$ moment conditions:

$$E(\boldsymbol{z_i}\varepsilon_i) = \mathbf{0}$$
$$\varepsilon_i = y_i - x_i'\beta$$

(a) write the relevant quadratic form to be minimized in the GMM framework;
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(c) write the expression for the *optimal* GMM estimator for β;
(d) show that the GMM estimator derived in (b) does not depend on the choice of the weighting matrix when R = K
(e) state under which conditions the GMM estimator derived in (b) is the OLS estimator.

*Answers:*
(a) *Slides part2 set 3 p.7*
(b) *Slides part2 set 3 p.8-9*
(c) *Slides part2 set 3 p.11-12*
(d) *Slides part2 set 3 p.20 (eq. 24)*
(e) *Slides part2 set 3 p.21 Interpretation: there are no endogenous variables in matrix X, so Z=X in this case*

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Where the variable "ed76" is education, exp and exp2 are experience and experience squared (over 100), black is a dummy variable equal to one if the person is black, reg76r is a dummy for living in the south and smsa76r is a dummy for urban area. Finally, nearc4 is the instrument and it is a dummy = 1 if the person grew up near 4-yr college.

a - what is considered the endogenous variable in this setting? What is the exogenous instrument? What are the other variables included in the first stage regression?

b – How can we test if the instrument is weak? What are the results in the example?

c – how can we test is the instrument is really exogenous, i.e. if it does not enter the main equation explaining earnings as a function of education and other covariates?

*Answers:*

*a- education is the endogenous variable, nearc4 is the instrument: to be a valid instrument it should have an effect on education but not a direct effect on earnings (in other words, it affects earnings only through its effect on education). The other covariates in the regression are considered exogenous, and they enter both the main regression and the first stage*

*b- Because there is just one instrument, we can test whether the instrument is weak simply looking at its t-statistic in the first stage (or, taking t^2, we have the F-test). The t-statistic of 4.18 tells us that the instrument is highly significant, and the F test is greater than 10 (conventional critical value – see slides).*

*c- the only way to test if the exogeneity assumptions are valid is to run an overidentifying restriction test, which is not possible here because the model is just identified R=K (so the Sargan statistic is equal to zero and has zero degrees of frredom).*

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```

In your model, you considered education as the only endogenous variable.

Explain what is a test of overidentifying restrictions in this context and how you would interpret it. How many instruments did you use in the estimation of your model?

*Answer: The Sargan statistic of overidentifying restrictions is the quantity ξ in the slides part 2 set 3, p. 26, (last line of the slide), which is a quadratic form in the moment restrictions that are minimized to find the estimator for our model. If the moment conditions are valid, this quadratic form will be small. It is distributed as a Chi^2 distribution with R-K degrees of freedom, where R is the number of moment conditions and K is the number of parameters to be estimated.*

1) Consider the problem where we estimate the K-dimensional parameter vector, θ by maximizing the likelihood function

$$\max_{\theta} \log L(\theta) = \max_{\theta} \sum_{i} \log L_i(\theta)$$

a) describe how we can obtain an estimator of the unknown parameter

b) state the properties of the estimator in point a), assuming the likelihood function is correctly specified

2) Consider the problem where we estimate the K-dimensional parameter vector, θ by maximizing the likelihood function

$$\max_{\theta} \log L(\theta) = \max_{\theta} \sum_{i} \log L_i(\theta)$$

We are interested in testing a set of J linear restrictions on the same parameter vector:

$$H_0: \boldsymbol{R}\theta = r$$

where **R** is a J×K matrix and r is a J×1 vector.

a) Broadly describe the three different principles that can be applied to test the null, and discuss advantages and disadvantages of each of them.
b) Consider in particular the Wald test and write down its formula. Explain briefly.
c) Assume that k=3 and that you want to test jointly whether $\theta_1 = \theta_2$ and $\theta_3 = 1$. Show how to write the elements of the matrix **R** and of the vector r in this case.

## Question 3.

We want to study how the probability of drinking varies with gender and age, on a sample of American individuals. The estimation results are as follow:

```
Logistic regression                                     Number of obs = 10,843
                                                        LR chi2(2)    = 554.51
                                                        Prob > chi2   = 0.0000
Log likelihood = -7091.6056                             Pseudo R2     = 0.0376

------------------------------------------------------------------------------
      rdrink | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    ragender |
   2.female |  -.6991671   .0404696   -17.28   0.000     -.778486   -.6198482
         age |   -.024887    .001859   -13.39   0.000    -.0285306   -.0212434
       _cons |   2.341921   .1291706    18.13   0.000     2.088751    2.595091
------------------------------------------------------------------------------
```

```
Average marginal effects                                  Number of obs = 10,843
Model VCE: OIM

Expression: Pr(rdrink), predict()
dy/dx wrt:  2.ragender age
```

| | dy/dx | Delta-method std. err. | z | P>\|z\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| ragender | | | | | | |
| 2.female | -.1668034 | .0095985 | -17.38 | 0.000 | -.1856161 | -.1479906 |
| age | -.0057471 | .0004157 | -13.83 | 0.000 | -.0065618 | -.0049324 |

Note: dy/dx for factor levels is the discrete change from the base level.

A) Explain what model has been used and why, and write down the likelihood and the log likelihood for the model in the previous point. Why don't we use a linear regression model?

B) Explain how we can interpret the coefficients reported in the first table.

C) Explain how we can interpret the marginal effects reported in the second table. Write down a formula for the marginal effects, and discuss if the marginal effects depend on the Xs and what does this mean.

**Question 4.**

We want to study how the probability of choosing to work or not to work varies with some individual characteristics. Briefly sketch which model you would use and how you would interpret the results *(hint: is it ok to interpret directly the coefficients?).*

1) Consider the problem where we estimate the K-dimensional parameter vector, θ by maximizing the likelihood function

$$\max_{\theta} \log L(\theta) = \max_{\theta} \sum_{i} \log L_i(\theta)$$

a) describe how we can obtain an estimator of the unknown parameter

*Answer: slides part3 set1, pp 14-16*

b) state the properties of the estimator in point a), assuming the likelihood function is correctly specified

*Answer: slides part3 set1, pp 17-22*

2) Consider the problem where we estimate the K-dimensional parameter vector, θ by maximizing the likelihood function

$$\max_{\theta} \log L(\theta) = \max_{\theta} \sum_{i} \log L_i(\theta)$$

We are interested in testing a set of J linear restrictions on the same parameter vector:

$$H_0: \boldsymbol{R}\theta = r$$

where **R** is a J×K matrix and r is a J×1 vector.

a) Broadly describe the three different principles that can be applied to test the null, and discuss advantages and disadvantages of each of them.

*Answer: slides part3 set2, pp 2-6*

b) Consider in particular the Wald test and write down its formula. Explain briefly.

*Answer: slides part3 set2, pp* 7-8

c) Assume that k=3 and that you want to test jointly whether $\theta_1 = \theta_2$ and $\theta_3 = 1$. Show how to write the elements of the matrix **R** and of the vector r in this case.

$$\boldsymbol{R} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad r = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

## Question 3.

We want to study how the probability of drinking varies with gender and age, on a sample of American individuals. The estimation results are as follow:

```
Logistic regression                                     Number of obs = 10,843
                                                        LR chi2(2)    =  554.51
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Log likelihood = -7091.6056                             Pseudo R2     =  0.0376
```

| rdrink | Coefficient | Std. err. | z | P>\|z\| | [95% conf. | interval] |
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```
Average marginal effects                                Number of obs = 10,843
Model VCE: OIM

Expression: Pr(rdrink), predict()
dy/dx wrt:  2.ragender age
```

| | Delta-method dy/dx | std. err. | z | P>\|z\| | [95% conf. | interval] |
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| age | -.0057471 | .0004157 | -13.83 | 0.000 | -.0065618 | -.0049324 |

```
Note: dy/dx for factor levels is the discrete change from the base level.
```

A) Explain what model has been used and why, and write down the likelihood and the log likelihood for the model in the previous point. Why don't we use a linear regression model?

*In general, the likelihood and log likelihood are those reported in equations 19, 20 and 21 of the slides (part4_1). The specific functional form for the logit is given by equations 9 and 10.*

B) Explain how we can interpret the coefficients reported in the first table.

*In general, in non linear models, the coefficients in the first table do not have a direct interpretation. However, in the simple logit and probit model, the sign of the coefficient is informative, because a positive coefficient on some variable implies an increase in the probability of Y=1 as a consequence of a unit increase in that variable, holding everything else constant.*

*In the logit model, additionally, we can interpret the coefficients as variations in the log odds of the dependent variable, hence exp(coefficient) will be informative on the variation in the odds ratio. For example, an increase in age by one unit will decrease the odds ratio of drinking by exp(-0.025)-1=-0.025 or -2.5%.*

C) Explain how we can interpret the marginal effects reported in the second table. Write down a formula for the marginal effects, and discuss if the marginal effects depend on the Xs and what does this mean.

*One way to interpret the parameters is to consider the marginal effects of changes in the explanatory variable. This is to be interpreted as a change in the probability (not on the odds ratio) in response to a marginal increase in a regressor. For a continuos variable, xk the marginal effect is defined as the partial derivative of the probability that yi = 1.*

*Formula (11) slides 4_1. No need to write more specific formula, but important to notice that the derivative here depends on the values of the explanatory variables. Solution: compute average marginal effects*

**Question 4.**

We want to study how the probability of choosing to work or not to work varies with some individual characteristics. Briefly sketch which model you would use and how you would interpret the results *(hint: is it ok to interpret directly the coefficients?)*.

*Answer: Because the dependent variable is binary, a logit or probit model would seem appropriate. The model is described in slides 8-11 of part3 set3.*

*To interpret the results (of any non linear model!) it is best to compute the marginal effects and not to interpret the estimated coefficients.*

**Question 1**

David Neumark and William Wascher published a study in 1992 of the effect of minimum wages on teenage employment using a U.S. state panel. The paper used annual observations for the years 1977-1989 and included all 50 states plus the District of Columbia. The estimated equation is of the following type:

$$Eit = \beta 0 + \beta 1\ (Mit\ /Wit\ ) + \gamma 2D2i + \ldots + \gamma nD51i + \delta 2B2t + \ldots + \delta TB13t + uit$$

where E is the employment to population ratio of teenagers, M is the nominal minimum wage, and W is average wage in the state. In addition, other explanatory variables, such as the prime-age male unemployment rate, and the teenage population share were included. D2-D51 are the state dummies, B2-B13 are the time dummies.

(a) Briefly discuss the advantage of using panel data in this situation rather than pure cross-sections or time series.

(b) Estimating the model by OLS but including only time dummy variables results in the following output

$$Eit = \beta 0 - 0.33 \times (Mit\ /Wit\ ) + 0.35 \times (SHY\ it\ ) - 1.53 \times uram\ it\ ; \quad =.20$$
$$\qquad\quad (.08) \qquad\qquad (.28) \qquad\qquad (.13)$$

where SHY is the proportion of teenagers in the population, and uram is the prime-age male unemployment rate. Coefficients for the time fixed effects are not reported.
Numbers in parentheses are homoskedasticity-only standard errors.

Comment on the above results. Are the coefficients statistically significant? Since these are level regressions, how would you calculate elasticities?

(c) Adding state fixed effects changes the above equation as follows:

$$Eit = \beta 0 + 0.07 \times (Mit\ /Wit\ ) - 0.19 \times (SHY\ it\ ) - 0.54 \times uram\ it\ ; \ = 0.69$$
$$(0.10) \qquad\qquad (0.22) \qquad\qquad (0.11)$$

Compare the two results. Why would the inclusion of state dummy variables change the coefficients in this way?

**Question 2**

Consider a simple regression problem for two time periods for the years 1982 and 1988 as follows:

$$FatalityRate_{i,1982} = \beta_0 + \beta_1 BeerTax_{i,1982} + u_{i,1982}$$
$$FatalityRate_{i,1988} = \beta_0 + \beta_1 BeerTax_{i,1988} + u_{i,1988}$$

Where FatalityRate_i is the number of annual traffic deaths every 10000 people in state *i* and BeerTax_i is the tax on a case of beer in 1988 U.S. dollars. After subtracting the first equation from the second equation, the authors estimate the model and find a negative intercept.

a. Show how you would have to modify the two equations to allow for the presence of an intercept

in the differenced model.
b. What would the relative magnitude of the modified model have to be for you to find a negative intercept?

**Question 3**

A major motivation for using panel data has been the ability to control for possibly correlated, time-invariant heterogeneity without observing it. Suppose a cross-sectional regression of the form

$y_{i1} = \beta x_{i1} + \alpha_i + u_{i1}$ with $E(u_{i1}|x_{i1}, \alpha_i) = 0$

a) State under which conditions $\beta$ can be consistently estimated if only one cross-section of data is available
b) Suppose T=2, so that we observe $y_{i2}$ and $x_{i2}$ for the same individuals, but the $\alpha_i$ are unobserved. How can $\beta$ be identified?
c) What assumptions do you need to make on the correlation between the regressor *x* and the individual unobserved effect for the estimator of $\beta$ to be consistent?

**Question 4**

1) Write down the linear Fixed effect model and specify the hypotheses that underlie the model.
2) Show how you can estimate the linear fixed effect model with the LSDV estimator, and discuss the properties of the estimator
3) Show how you can estimate the linear fixed effect model with the within estimator, and discuss the properties of the estimator.
4) Show how you can estimate the linear fixed effect model with the first difference estimator, and discuss the properties of the estimator.

**Question 5**

Consider the model:

$y_{it} = \beta x_{it} + \alpha_i + u_{it}$ with $E(u_{it}|x_{it}, \alpha_i) = 0$ and T=2. Show that the FE and the FD estimators coincide.

**Question 1**

David Neumark and William Wascher published a study in 1992 of the effect of minimum wages on teenage employment using a U.S. state panel. The paper used annual observations for the years 1977-1989 and included all 50 states plus the District of Columbia. The estimated equation is of the following type:

$$Eit = \beta 0 + \beta 1\ (Mit\ /Wit\ ) + \gamma 2D2i + \ldots + \gamma nD51i + \delta 2B2t + \ldots + \delta TB13t + uit$$

where E is the employment to population ratio of teenagers, M is the nominal minimum wage, and W is average wage in the state. In addition, other explanatory variables, such as the prime-age male unemployment rate, and the teenage population share were included. D2-D51 are the state dummies, B2-B13 are the time dummies.

(a) Briefly discuss the advantage of using panel data in this situation rather than pure cross-sections or time series.

(b) Estimating the model by OLS but including only time dummy variables results in the following output

$$Eit = \beta 0 - 0.33 \times (Mit\ /Wit\ ) + 0.35 \times (SHY\ it\ ) - 1.53 \times uram\ it\ ; \quad =.20$$
$$\qquad\quad (.08) \qquad\qquad (.28) \qquad\qquad (.13)$$

where SHY is the proportion of teenagers in the population, and uram is the prime-age male unemployment rate. Coefficients for the time fixed effects are not reported.
Numbers in parentheses are homoskedasticity-only standard errors.

Comment on the above results. Are the coefficients statistically significant? Since these are level regressions, how would you calculate elasticities?

(c) Adding state fixed effects changes the above equation as follows:

$$Eit = \beta 0 + 0.07 \times (Mit\ /Wit\ ) - 0.19 \times (SHY\ it\ ) - 0.54 \times uram\ it\ ; \ = 0.69$$
$$(0.10) \qquad\qquad (0.22) \qquad\qquad (0.11)$$

Compare the two results. Why would the inclusion of state dummy variables change the coefficients in this way?
Answer:
(a) There are likely to be omitted variables in the above regression. One way to deal with some of these is to introduce state and time effects. State effects will capture the influence of omitted variables that are state specific and do not vary over time, while time effects capture those of country wide variables that are common to all states at a point in time. Furthermore, there are more observations when using panel data, resulting in more variation.
(b) There is negative relationship between minimum wages and the employment to population ratio. Increases in the share of teenagers in the population result in a higher employment to population ratio, and increases in the prime-age male unemployment rate lower the employment to population ratio. 20 percent of employment to population of teenagers variation is explained by the above regression. The relative minimum wage and the prime-age male unemployment rate are significant using a 1% significance level, while the proportion of teenagers in the population is not. A 0.01 increase in $(M_{it}/W_{it})$ results in a 0.33*0.01=0.0033 decrease in E, that is a 0.33 percentage points

reduction in the employment rate of teenagers.
(c) The parameter of interest here is the coefficient on the relative minimum wage. While it was highly significant in the previous regression, it now has changed signs and is statistically insignificant. The explanatory power of the equation has increased substantially. The size of the other two coefficients has also decreased. The results suggest that omitted variables, which are now captured by state fixed effects, were correlated with the regressors and caused omitted variable bias.

**Question 2**

Consider a simple regression problem for two time periods for the years 1982 and 1988 as follows:

$$FatalityRate_{i,1982} = \beta_0 + \beta_1 BeerTax_{i,1982} + u_{i,1982}$$
$$FatalityRate_{i,1988} = \beta_0 + \beta_1 BeerTax_{i,1988} + u_{i,1988}$$

Where FatalityRate_i is the number of annual traffic deaths every 10000 people in state *i* and BeerTax_i is the tax on a case of beer in 1988 U.S. dollars. After subtracting the first equation from the second equation, the authors estimate the model and find a negative intercept.

a. Show how you would have to modify the two equations to allow for the presence of an intercept in the differenced model.
b. What would the relative magnitude of the modified model have to be for you to find a negative intercept?
Answer:
a. $FatalityRate_{i,1982} = \beta_0 + \beta_1 BeerTax_{i,1982} + u_{i,1982}$
$FatalityRate_{i,1988} = \alpha_0 + \beta_1 BeerTax_{i,1988} + u_{i,1988}$

b. $\alpha_0 < \beta_0$

**Question 3**

A major motivation for using panel data has been the ability to control for possibly correlated, time-invariant heterogeneity without observing it. Suppose a cross-sectional regression of the form

$y_{i1} = \beta x_{i1} + \alpha_i + u_{i1}$ with $E(u_{i1}|x_{i1}, \alpha_i) = 0$

a) State under which conditions $\beta$ can be consistently estimated if only one cross-section of data is available
b) Suppose T=2, so that we observe $y_{i2}$ and $x_{i2}$ for the same individuals, but the $\alpha_i$ are unobserved. How can $\beta$ be identified?
c) What assumptions do you need to make on the correlation between the regressor *x* and the individual unobserved effect for the estimator of $\beta$ to be consistent?

*Answers:*

*a) Slides part_4 set 1, p. 5-6: there are 3 cases: i) the $\alpha_i$ are observed, 2) the $\alpha_i$ are orthogonal to the regressor(s), iii) an external instrument is available (equation 3 in the slides)*
*b) Slide p.7*
*c) For consistency of the estimator in point b) we do not need to assume orthogonality between the unobserved $\alpha_i$ and the regressor(s), hence correlation may be different from zero*

**Question 4**

1) Write down the linear Fixed effect model and specify the hypotheses that underlie the model.
2) Show how you can estimate the linear fixed effect model with the LSDV estimator, and discuss the properties of the estimator
3) Show how you can estimate the linear fixed effect model with the within estimator, and discuss the properties of the estimator.
4) Show how you can estimate the linear fixed effect model with the first difference estimator, and discuss the properties of the estimator.

*Answers:*

1-4) see slides part4 pp. 8-15

**Question 5**

Consider the model:

$y_{it} = \beta x_{it} + \alpha_i + u_{it}$ with $E(u_{it}|x_{it}, \alpha_i) = 0$ and T=2. Show that the FE and the FD estimators coincide.

*Answer:*

**First Difference (FD) Estimator**

Compute differences:

$$\Delta y_i = y_{i2} - y_{i1}$$
$$\Delta x_i = x_{i2} - x_{i1}$$
$$\Delta u_i = u_i2 - u_{i1}$$

FD regression:

$$\Delta y_i = \beta \Delta x_i + \Delta u_i$$

FD estimator:

$$\hat{\beta}_{FD} = ( \Sigma \Delta x_i \Delta y_i ) / ( \Sigma (\Delta x_i)^2 )$$

**Fixed Effects (FE) Estimator When T = 2**

Compute individual means:

$$\bar{y}_i = (y_{i1} + y_{i2})/2$$
$$\bar{x}_i = (x_{i1} + x_{i2})/2$$

Within-transformed values:

For t = 1:

$$y_{i1} - \bar{y}_i = -½ \Delta y_i$$
$$x_{i1} - \bar{x}_i = -½ \Delta x_i$$

For t = 2:

$$y_{i2} - \bar{y}_i = ½ \Delta y_i$$
$$x_{i2} - \bar{x}_i = ½ \Delta x_i$$

**FE Estimator Calculation**

The FE estimator is:

$$\hat{\beta}_{FE} = [ \Sigma_t \Sigma_i (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i) ] / [ \Sigma_t \Sigma_i (x_{it} - \bar{x}_i)^2 ]$$

Substitute the within-transformed values:

Numerator for each i:

$$(-½ \Delta x_i)(-½ \Delta y_i) + (½ \Delta x_i)(½ \Delta y_i)$$
$$= ¼ \Delta x_i \Delta y_i + ¼ \Delta x_i \Delta y_i = ½ \Delta x_i \Delta y_i$$

Denominator for each i:

$$(-½ \Delta x_i)^2 + (½ \Delta x_i)^2$$
$$= ¼ (\Delta x_i)^2 + ¼ (\Delta x_i)^2 = ½ (\Delta x_i)^2$$

Thus:

$$\hat{\beta}_{FE} = ( \Sigma ½ \Delta x_i \Delta y_i ) / ( \Sigma ½ (\Delta x_i)^2 )$$
$$= ( \Sigma \Delta x_i \Delta y_i ) / ( \Sigma (\Delta x_i)^2 )$$

Therefore, FE and FD estimators are exactly identical when T = 2.